EWB VPC Peering Technical Implementation Report

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1. Executive Summary & Infrastructure Overview

This project demonstrates secure cross-VPC networking within AWS using a VPC Peering Connection (pcx-007da753d9e0e43f3). Network traffic was routed between VPC A (10.10.0.0/16) and VPC B (10.20.0.0/16) without traversing the public internet. Testing validated both baseline operational connectivity and security controls via controlled failure and recovery scenarios.

Resource Details:

- VPC A ID: vpc-05ad55eda3a7e90ef (10.10.0.0/16)
- VPC B ID: vpc-0fa91f7c50562bee8 (10.20.0.0/16)
- Bastion Host (ewb-bastion): ID i-0b4eb2e6f16fae14a | Private IP: 10.10.1.50
- App Server A (ewb-app-a): ID i-0d449c54c91b95d67 | Private IP: 10.10.2.17
- App Server B (ewb-app-b): ID i-0d6bdbd943e74893a | Private IP: 10.20.1.48

2. Core Infrastructure Setup Evidence

- EC2 Instance Inventory: All required instances (ewb-bastion, ewb-app-a, and ewb-app-b) are active with healthy status checks.
- VPC Peering Connection: VPC Peering connection pcx-007da753d9e0e43f3 is Active between Requester VPC (vpc-05ad55eda3a7e90ef) and Accepter VPC (vpc-0fa91f7c50562bee8).

3. Routing & Security Controls Evidence

- VPC A Route Table (ewb-rt-a-public): Route for 10.20.0.0/16 targets pcx-007da753d9e0e43f3.
- VPC B Route Table (ewb-rt0b-private): Route for 10.10.0.0/16 targets pcx-007da753d9e0e43f3.

- Security Group Inbound Rules (ewb-sg-app-b): Security group sg-0b7d06fc139d99c64 allows inbound SSH (TCP 22) and Custom ICMP (Echo Request) traffic strictly from VPC A (10.10.0.0/16).

4. Testing & Validation Evidence

- Controlled Failure Scenario (100% Packet Loss): When the ICMP inbound rule was temporarily removed from ewb-sg-app-b, cross-VPC ping attempts from ewb-bastion (10.10.1.50) to ewb-app-b (10.20.1.48) failed completely with 100% packet loss.

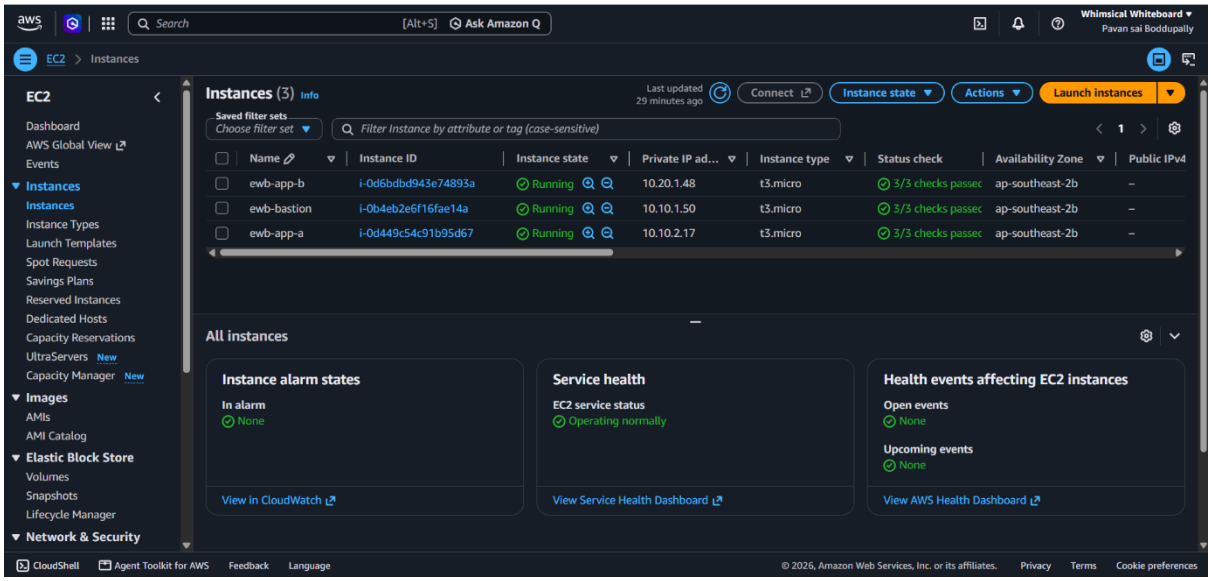
- Recovery Verification (0% Packet Loss): Upon re-enabling the ICMP rule on ewb-sg-app-b, end-to-end network connectivity was successfully restored with 0% packet loss.

5. Conclusion

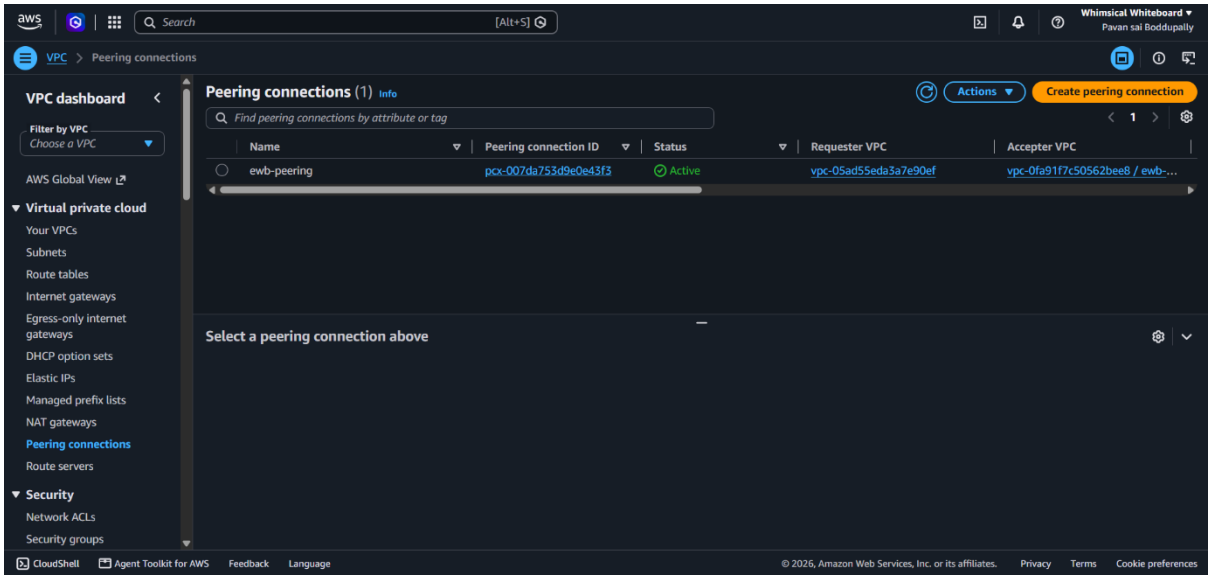
The Enterprise Web Bastion (EWB) VPC Peering architecture has been deployed, configured, and verified according to all technical requirements.

6.screen shorts of project

- core infrastructure setup evidence



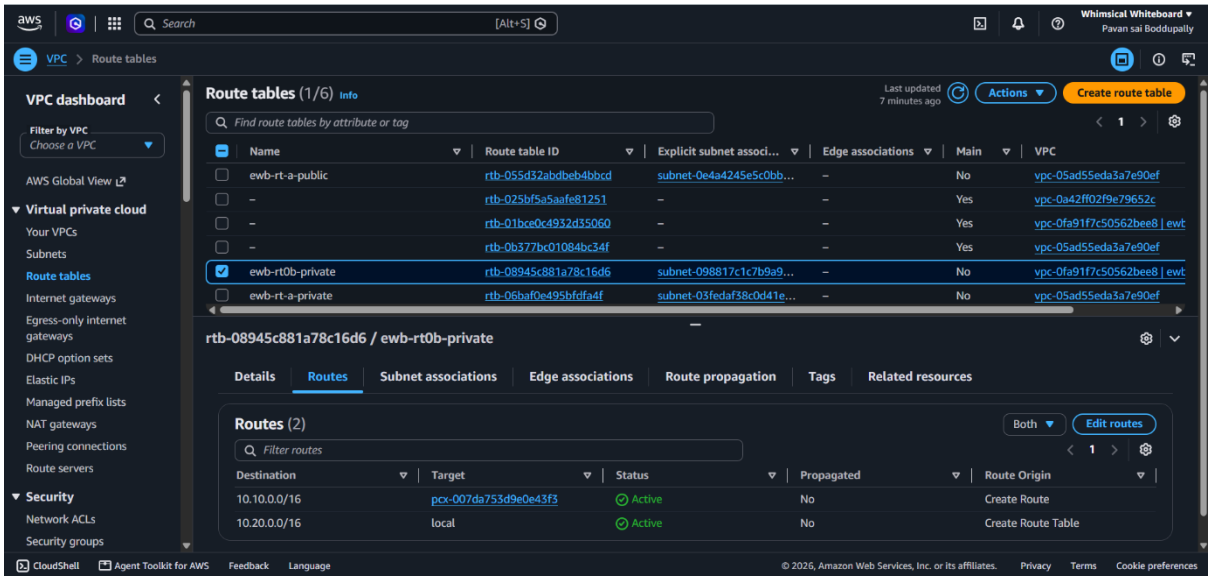
EC2 instances screen short



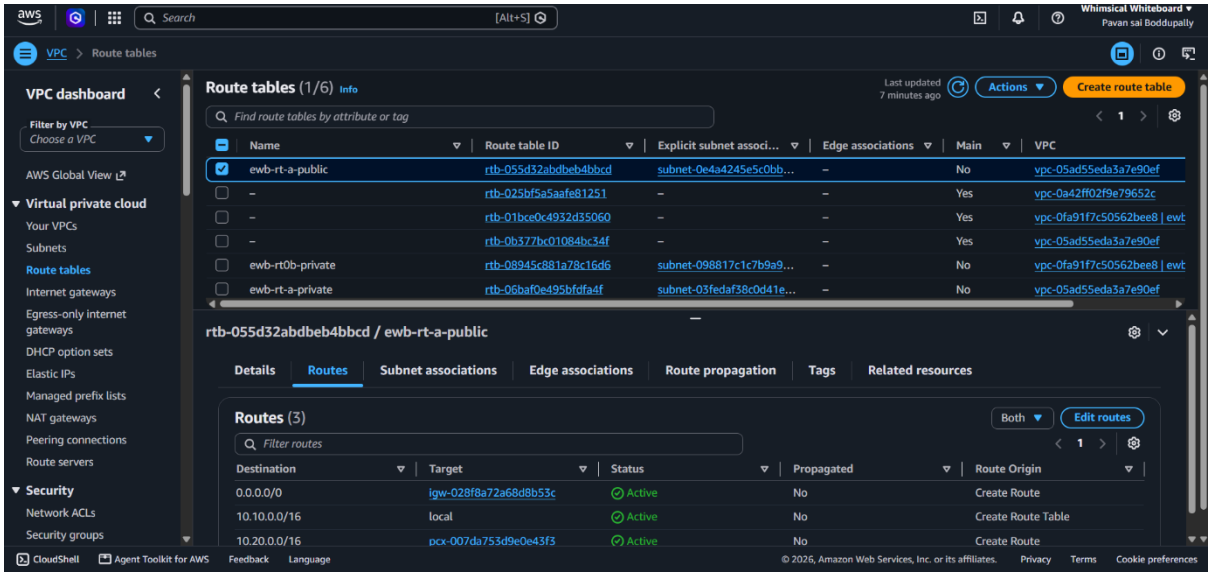
VPC peering screenshot

-routing and security controls evidence

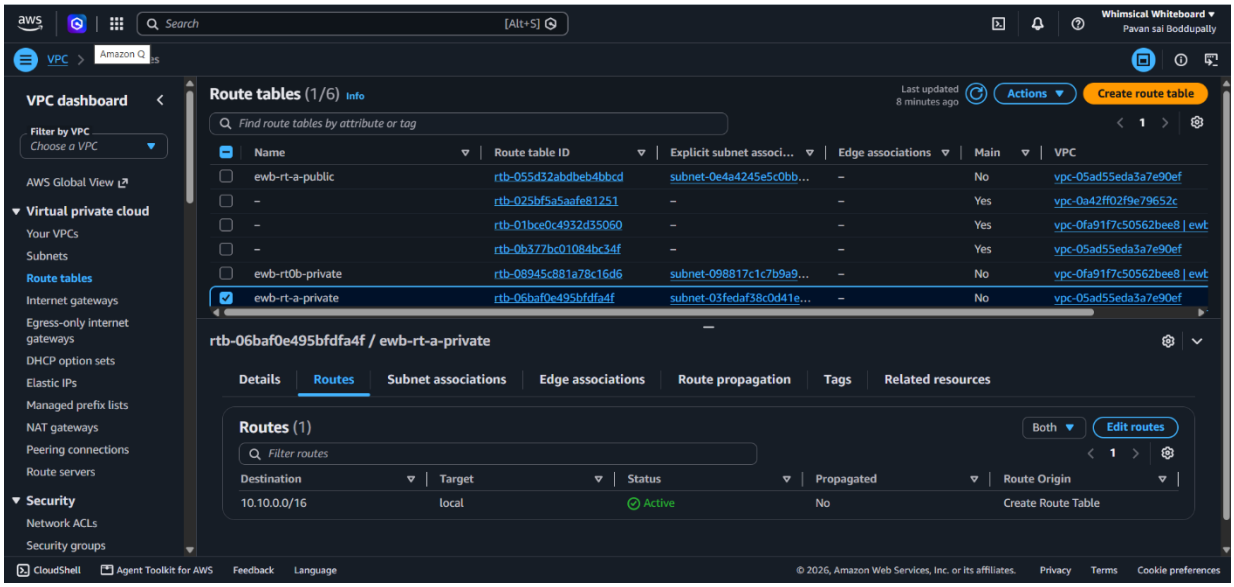
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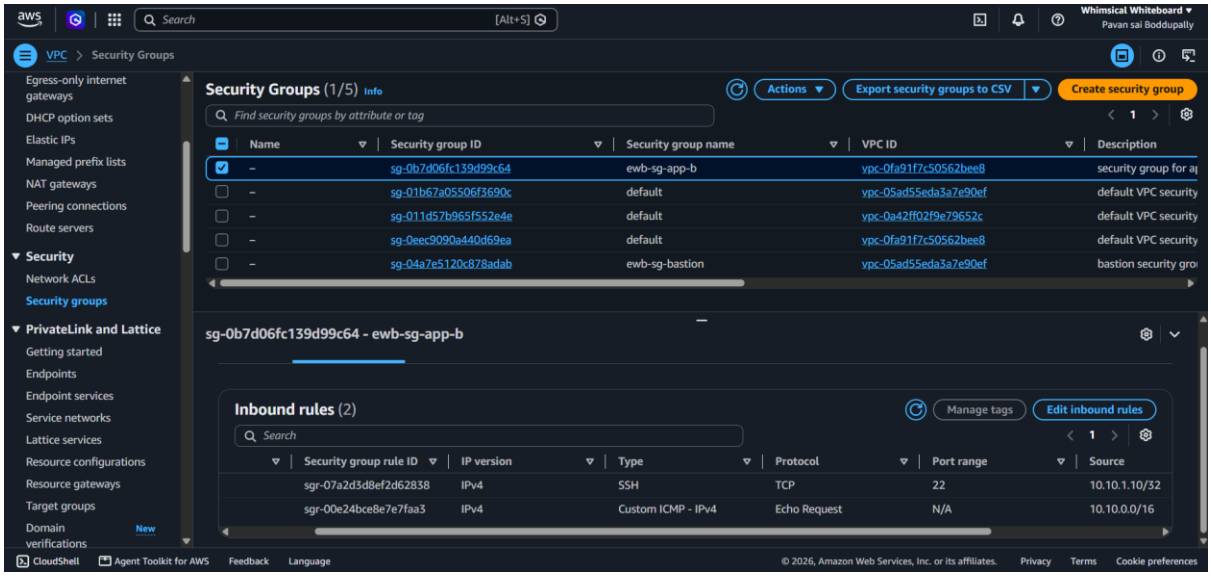
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Routing table screenshot



Security group screenshot

-testing and validation evidence



Failure ping screenshot



Recovery ping screenshot